TAQI EDDINE EL MAMOUNI

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SUMMARY

AI Engineer with hands-on experience in developing intelligent systems using Deep Learning, Natural Language Processing (NLP), and Computer Vision. Specialized in building end-to-end AI solutions for real-time applications in physiotherapy, robotics, and data analysis. Proficient in model engineering, fine-tuning LLMs, and deploying AI models via scalable APIs. Passionate about applied research, innovation, and transforming advanced AI concepts into real-world technologies.

EXPERIENCE

Data Analyst

Nüfus ve Sosyal Arat rmalar Enstitüsü (NSA) **October 2025 - Present, Istanbul, Turkey**

- Performed statistical and predictive data analysis for social and demographic datasets using Python (Pandas, NumPy, Matplotlib).
- Built Power BI dashboards and automated data cleaning pipelines.
- Improved dataset quality through custom-built validation scripts.

Al & Computer Vision Engineer (Internship + Full-Time)

Deeptech Mühendislik Tic. Ltd. ti. **June 2025 - Present, Istanbul, Turkey**

- Developed advanced AI-powered pose estimation systems using MediaPipe, YOLO, and OpenCV for human posture and movement analysis.
- Designed real-time AI models for physiotherapy and ergonomic correction applications.
- Integrated inference pipelines into Flask APIs and built interactive dashboards for visual posture tracking.
- Implemented 3D skeleton visualization in Flutter using CustomPainter.

Computer Vision Trainee

Alnahda Initiative (Nwat) **August 2025 - October 2025, Istanbul**

- Used OpenCV to track a ball in real time.
- Controlled a 3RRS robotic platform with an STM32 microcontroller via PID & inverse kinematics.
- Integrated vision and embedded systems for fully autonomous operation.

Machine Learning Intern

Mostaql.com **March 2024 - February 2025, Remote - Morocco**

- Created ML pipelines for image and text classification projects using TensorFlow and scikit-learn.
- Improved model accuracy by over 10%.
- Delivered reports explaining model behavior and results.

Web Developer

Prestij Consulting / Core Istanbul **January 2023 - August 2023, Istanbul, Turkey**

- Built responsive front-end interfaces using React.js and Tailwind CSS.
- Integrated RESTful APIs and optimized web performance.
- Enhanced UX/UI design for mobile and desktop.

PROJECT

Real-Time Ball Balancing Robot

- Vision-guided robotic control using OpenCV and STM32.

Pose Estimation using YOLO & MediaPipe

Deeptech Mühendislik Tic. Ltd. Şti

- Real-time 33-keypoint tracking; 3D Flutter viewer.
- Designed a real-time tracking system for physiotherapy and ergonomic assessment.

Potato Leaf Disease Classification CNN model in TensorFlow

iskenderun teknik üniversitesi

- achieving 93% accuracy.
- Developed a Potato Disease Classification system using CNN and TensorFlow to detect leaf diseases.
- Deployed with a React Native frontend and Google Cloud Functions for real-time predictions.
- A step toward smarter, AI-powered agriculture!

Medical Chatbot

iskenderun teknik üniversitesi
- Promotes Turkish-language NLP applications
- Real-world health chatbot use-case using LLaMA 3
- End-to-end AI integration (data, training, deployment)
- Fully functional Flask web app with real-time responses

Automated News Summarizer
iskenderun teknik üniversitesi
- Fine-tuned LLM summarizing long articles.

EDUCATION

Bachelor of Science in Computer Engineering
Iskenderun Technical University (ISTE) • Iskenderun • 2021 – 2025

CERTIFICATIONS

Machine Learning with Python
freeCodeCamp • 2025

Large Language Models (LLMs)
freeCodeCamp • 2025

Al Agents Fundamentals
Hugging Face • 2025

NLP
Bilgi Teknolojileri ve İletişim Kurumu - BTK - ICT • 2024

Deep Learning
BTK Akademi • 2023

INVOLVEMENT

Computer Vision
ROSAT • September 2025 - Present
- The DEPRON project develops an intelligent, AI-powered system for real-time disaster area analysis and humanitarian aid coordination.
- It uses Unmanned Aerial Vehicles (UAVs) to capture high-resolution video over zones like earthquakes and floods.
- The data is instantly processed to detect human density and priority areas through advanced tracking and counting algorithms.
- Any zone detecting 100 or more people is flagged as a critical humanitarian aid area for rapid emergency response.

Al & NLP Developer (Member)
skenderun • Al Club - Iskenderun Technical University (ISTE) • October 2024 - Present
- Created educational NLP & LLM projects in Arabic and Turkish.
- Fine-tuned transformer chatbots and led Deep Learning workshops

SKILLS

Languages : Python, JavaScript

Frameworks & Tools : React, NodeJS, FastAPI,WebRTC.

Computer Vision : YOLO, MediaPipe, Object Detection, Image Analysis,OpenCV .

AI/ML/DL : TensorFlow, PyTorch, scikit-learn, Keras, CNNs, RNNs, LSTM, NLP, Pandas, NumPy, Matplotlib, Seaborn.

Databases : MySQL, MongoDB, Firebase

SOther Tools: Git, VS Code, Figma, Power BI, Photoshop,Figma, Adobe XD, Photoshop, Illustrator, Adobe Premiere.

LANGUAGES

Arabic, English, Turkish.